Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning Engineer with 3 years of industry experience building and operating production **Python** and **SQL** data pipelines, **MLOps** deployment workflows and **GenAI** systems on **AWS**, including a scheduled pipeline that saves **60 analyst hours a month** and **RAG** services over **PostgreSQL/pgvector** shipped through one-click **Jenkins and GitLab CI/CD**. Rebuilt a parsing pipeline when its input source changed format and guardrailed LLM behaviour with versioned prompts. On an MSc in Data Science (University of Bristol, awaiting result), ran multi-seed CNN/ViT experiments, co-built an evidence-backed scoring and ranking tool with AstraZeneca and placed 3rd in a Lloyd's Bank datathon. Worked directly with business teams to turn needs into working tools. Seeking to join ATG's ML team to productionise models, build data pipelines and MLOps workflows, and help design LLM-powered decision-support tools.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Data Pipelines & Automation

- Built a **Python** pipeline, scheduled at 9 AM through **EventBridge** and **Lambda** and run on **SageMaker**, that parses over **300 daily emails** from credit rating agencies, fuzzy-matches company names against the ones analysts track and sends a consolidated daily summary, saving **60 analyst hours a month**; parsed first with **AWS Textract**, then rebuilt on **Azure OpenAI** when an agency began sending emails as images.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the licensed **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity) and integrated with legacy analyst data; developed **Optimized SQL** queries, **Functions** and **Procedures** for preprocessing and data insertion.
- Analyzed customer data with **Python Pandas**, and ran **Market Basket Analysis** (**Apriori Algorithm**) to identify customer buying patterns, handing the insights to the marketing team.

MLOps, Deployment & Reliability

- Designed the infrastructure for internal analytics platforms and GenAI applications on **AWS** (**EC2** including a **GPU** instance, **S3**, **ECR** and **Lambda**); packaged the **FastAPI** microservices (**uvicorn**) as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Chose and operated a per-department model mix on quality and cost: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department's traffic, kept switchable to other models, with the stronger **Azure OpenAI** outputs for decision makers; released to real users and iterated on their feedback.

LLM Systems & Stakeholder Delivery

- Designed and built production **GenAI** applications end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & Llama-3) serving Compliance, Client Service and Investment, with a separate **pgvector** database per department.

- Built **tool-calling** agents that call **Python** functions to run **SQL**-backed fund comparisons and generate PDF reports on request; guardrailed behaviour with **system prompts**, versioned every prompt change in **Git** and had the **Compliance** department review each release.
- Migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**, then recreated the dashboards in the **ReactJS** frontend so the wider organization could use them at lower cost than Power BI licences.
- Worked directly with stakeholders in Client Service, Compliance and Investment to turn requirements into working tools; **de facto team lead** for up to **4 interns** in my final year; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca: CellLineSelector (**Team project, co-built**. Preprocessing pipeline turning 17 raw inputs into 14 clean joined tables with identity resolution and **integrity checks**, plus a **scoring and ranking** algorithm that ranks cancer cell lines by target inclusion and exclusion genes across six evidence layers and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- University of Bristol MSc Applied Data Science Group Project: VisionCaptioner (**Team project**. Trained SimpleCNN, ResNet-18 and a custom ViT encoder from scratch and compared SBERT embedding-retrieval and LSTM/attention decoders for image captioning on three synthetic datasets, across **five seeds**, evaluated with **SBERT cosine similarity**, BLEU and ROUGE-L against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- Wingman: AI dating-coach web app with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: **validated with Pydantic**, repaired when malformed, vision calls **retried once**, partial results salvaged and responses **cached by input hash**; calls serialised under rate limits and results streamed to the UI. GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- Project Hack 27: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- Encode Vibe Coding Hackathon: Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- University of Bristol Datathon sponsored by Lloyd's Bank: 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- Tools: Docker, Jenkins, GitLab (CI/CD), Git, AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker, Textract), Azure OpenAI, LlamaIndex, Power BI, Nginx, Linux, LDAP, VS Code, Claude Code.
- Languages: Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- Databases: PostgreSQL (schema design, pgvector semantic search, SQL functions and procedures), SQLite